

Bell doesn't play dice!
Study on the Classical Mechanical Origin of
Quantum Entanglement

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Abstract

Within the most general framework allowed by the classical Principle of Least Action, we find that a specific subclass of classical dynamics, characterized by Intrinsic Periodicity of spacetime, potentially leads to violations of Bell's inequalities. This violation of Bell's inequality at a classical level is coherent with the findings of our previous works where we have extensively proven the exact formal equivalence between the statistical description of these ultra-fast inherently classical cyclic dynamics and the predictions of standard quantum mechanics. We have a scenario concealing Einstein's vision of a causal, local reality with Bell's request of non-locality, without involving any hidden variable. In particular, these cyclic classical dynamics introduce a peculiar element that could be interpreted as a "non-locality" for what which concerns the Bell experiment, but it is in truth fundamentally distinct from the non-locality postulated in quantum mechanics. It is manifestly compatible with classical mechanics and is also implicitly present in General Relativity, without breaking causality. The apparent indeterminism of quantum mechanics seems to arise from the present fundamental experimental limitation in directly observing these ultra-fast cycles, which occur on timescales of approximately $\sim 10^{-21}$ seconds.

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1 Introduction

In 1935, Einstein, Podolsky, and Rosen (EPR) proposed a *Gedanken* experiment to argue for the incompleteness of the Copenhagen interpretation of Quantum Mechanics (QM), [1]. Standard QM postulates that conjugate variables are non-commuting, leading to the existence of entangled two-particle states. In these states, measuring a variable of one particle determines the corresponding conjugate variable of the distant second particle. EPR suggested that a classical theory might underlie QM, with QM emerging as a statistical approximation. In this paper we investigate the possibility of this scenario.

To explore this possibility, Bell (1964) introduced local hidden variables, [2, 3], in addition to the ordinary spacetime coordinates, assuming that these variables contain all the possible “elements of reality” suggested by EPR, from which QM arises statistically. However, Bell’s theorem demonstrated that the broad class of classical local Hidden Variable Theories (HVT) exhibit statistical correlations slightly weaker than those predicted by QM. These correlations are expressed as Bell’s inequalities, which constrain classical HVT as a signature of its non-local nature. Consequently, according to Bell, only non-local HVT can reconcile with QM predictions, but these theories conflict with classical-relativistic locality, potentially violating Lorentz invariance.

However, hidden variables was never mentioned by Einstein as solution of the incompleteness of QM, [3]. Bell’s theorem does not preclude the existence of a classical theory that completes QM if Local HVT (LHVT) do not cover all possible classical-relativistic theories compatible with the requirement of relativistic causality and locality¹. This paper explores such a possibility, applying the EPR argument to a classical-relativistic theory that extends beyond QM, [6–16]. This theory, formally proven to be statistically indistinguishable from standard QM, avoids hidden variables altogether — a basic hypothesis of Bell’s theorem. Instead, it is based on a non-trivial topology of relativistic spacetime, introducing a novel form of “non-locality” that, remarkably, remains consistent with classical-relativistic locality while still violating Bell’s inequalities.

Classical Mechanics (CM), renamed here for convenience General Classical Mechanics (Gen-CM), encompasses all the possible physical systems whose dynamics are governed by the classical Principle of Stationary Action (PSA). This means that they minimize the Action both in the ‘bulk’ and at the boundary of a given time interval between an initial time and a final time. While all systems within Gen-CM obey the Euler-Lagrange equations (EL eqs) as their Equations of Motion (EoMs) coming from the minimization in the ‘bulk’ of a time interval, the specific solutions are determined by the Boundary Conditions (BCs) chosen to minimize the Action at the boundaries. *Satisfying these BCs, required by the PSA for the boundary of the time interval, generally ensures full locality and causality for the resulting mechanics, without violating Lorentz invariance or other classical symmetries.* Importantly,

¹ On the contrary, Einstein appears to point toward an approach closely aligned with ECT. As reported by A. Pais, Einstein was convinced that “it is necessary to start from classical field theories and ask that quantum laws emerge from constraints imposed to them” [4]. Einstein wrote [5]: “*For sure; we must just overdetermine the variables of the matter wave or field by means of constraints. The dynamics of the particles would be overdetermined in such a way that the initial conditions would be subject to restrictive constraints*”. A.Einstein (1923). He further added a requirement of covariance for these constraints.

we distinguish two sets of BCs with the above compatibility with the PSA, with two resulting subclasses of Gen-CM, see fig.(1).

The first subclass of Gen-CM is nothing but the common Standard Classical Mechanics (Std-CM) — see Hamilton’s Principle [17]. As we know, the Std-CM are the class of synchronous varied motions that conserve the configurations of the system at the boundary of a time interval. They are the stationary solutions of the EL eqs characterized by vanishing variations at the time boundaries. These BCs will be termed in this paper Standard BCs (Std-BCs²).

The PSA equally allows another choice of BCs — often overlooked — minimizing the action at the boundary as much as for the Std-BCs. It involves combinations of Dirichlet or Neumann BCs, as well known from string or Kaluza-Klein theories [18–21]. For the sake of simplicity, among these BCs here we consider exclusively Periodic Boundary Conditions (PBCs) — which can be written as combination of two Dirichlet or Neumann BCs. The resulting particular subclass of Gen-CM will be referred to as Cyclic CM (Cyclic-CM). *It is important to notice that the adherence of these PBCs to the PSA guarantees the strict locality and causality of the resulting Cyclic-CM*, despite the fact that the PBCs implies an Intrinsic Periodicity (IP) on the coordinates, [18–21]. This is because these PBCs, and thus the IP on the coordinates, are dynamical functions as much as the periodicity of relativistic clocks in General Relativity (GR) or the frequency (and thus the periodicity) of the wave-functions on undulatory mechanics, as shown in [6–16].

The Cyclic-CM, derived by applying PBCs within the framework of PSA, were originally investigated in [16] and are described by the Elementary Space-Time Cycles Theory (ECT), extensively explored in [6–15]. In general, ECT is based on a non trivial classical topology of curved space-time (intrinsically cyclic space-time arising from the imposed PBCs) and the resulting cyclic dynamics are predicted to be ultra-fast with respect our experimental time resolution, being determined by the mass of the elementary particles. Since every detector and signal is eventually based on electromagnetism, inevitably involving internal dynamics of the electrons, they can only be directly observed at time resolutions better than the Compton time of the electron $\sim 10^{-21}$ seconds. It provides an alternative perspective of relativistic space-time, and of relativity itself, entirely consistent with all known physics as rigorously demonstrated in the previous work. Crucially, the IP inherent in these mechanics seems to offer a novel solution to the long-standing conflict between the non-locality inherent in Bell’s theorem and the principle of locality advocated by Einstein.

We have a fundamental and absolutely general theorem about Cyclic-CM, proven in [7]: as a matter of mathematical facts, the probabilistic description, classical and deterministic in the essence, resulting from the Cyclic-CM associated to any classical Hamiltonian (symplectic) system by imposing PBCs rather than Std-BCs, is fully equivalent to the expectation values of the standard QM resulting from the ordinary quantization of the classical system itself. The indistinguishability between the statistical outcomes predicted by Cyclic-CM and those predicted by standard QM has been rigorously proven for all major quantization procedures, [6–16]. It includes all the desired key quantum features such as commutation relations, Hilbert space and

²In [6, 7, 13–16] they were named Synchronous BCs

Schrödinger dynamics. They are also naturally described at a classical level by the ordinary Feynman Path Integral. This comprehensive body of evidence proves that Cyclic-CM is potentially the classical physics from which QM emerges. Consequently, the violation of Bell’s inequalities within the framework of Cyclic-CM, the central thesis of this paper, is an expected outcome.

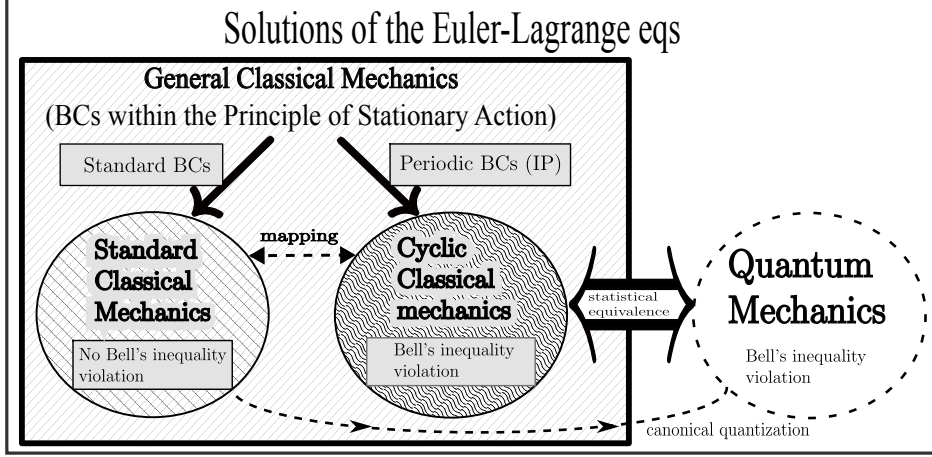


Fig. 1 Schematic representation of solutions to the Euler-Lagrange equations within a generic Hamiltonian system (large box). They minimize the action in the bulk of a time interval. Within the Principle of Stationary Action which also requires minimization at the boundary (small bold box), different BCs give rise to distinct subclasses of CM. Std-CM, *i.e.* vanishing variation of the solutions at the time boundaries, leads to the Standard CM, including HVT, which do not exhibit violations of Bell’s inequalities (left circle). Periodic BCs, equally minimizing the action at the boundary, give rise to Cyclic-CM, characterized by the violation of Bell’s inequalities (central circle). It is proven [6–16] that the statistical description of the Cyclic CM exhibit a remarkable full equivalence with the standard QM obtained by the quantization of the Hamilton dynamics itself, suggesting a classical foundation for quantum phenomena. The Std-CM and the Cyclic-CM can be mathematically related (mapping) through a suitable choice of the time interval over which the action is defined.

In this paper we exclusively focus on the fundamental properties of Cyclic-CM that give rise to violations of Bell’s inequality relevant to the CHSH experiment, [22, 23], demonstrating that such violations can emerge within a purely classical framework, Par.(3). For the purpose of this analysis, it will be sufficient to introduce basic aspects of ECT, leaving the details of the advanced aspects, such as local modulations of IP within curved spacetime, to previous work — only mentioned in Par.(2) and Par.(5).

In fact it will be sufficient to investigate dynamics of “signals” freely traveling at the speed of light and their interactions with polarizers. This translates to examining Cyclic dynamics characterized by persistent recurrences in time and space. The effect of the polarizers is given by simple phase shifts of these persistent recurrences. To avoid confusion with QM ‘photons’, we generically refer to these classical signals as “massless clocks”, or simply “clocks”. Remarkably, these “massless clocks” of Cyclic-CM exhibit dynamics that precisely mirror the behavior of ordinary quantum ‘photons’ in the CHSH experiment, despite their inherently classical nature.

In general, the violation of Bell’s inequality is primary a direct consequence of the non-commutativity postulated in the mathematical formulation of QM, as also originally highlighted by Einstein’s arguments concerning the incompleteness of QM, [1], see also [24]. This postulated non-commutativity lies at the heart of all non-local aspects of quantum phenomena. Contrarily, in [7], and in [6] specifically for the second quantization, and previous papers specifically for the Feynman Path Integral, we have demonstrated that the condition of IP associated to general Hamiltonian systems directly leads non-commutativity (identical to QM) at a classical level, without need to postulate them, as described in Par.(2) and Par.(3). This allows us to develop a formalism of Gen-CMs that enables dynamical analysis without pre-specifying the chosen BCs, Par.(3.1). Consequently, by subsequently imposing the appropriate BCs, we can separately derive results for both Std-CM and the Cyclic-CM. Essentially, these results can be directed extracted from the boundary and bulk terms of our Gen-CM expressions, respectively. In other words, Std-CM is the “holography projection” of the Cyclic-CM on the boundary of the theory (and therefore of the standard QM).

In Par.(3.2) we also develop a model that effectively mirror the generation of quantum entangled states at a classical level. This model is based on Cyclic dynamics defined on a two site lattice, essentially an ultra-fast “tic-tac clock” mechanism. It is developed by using ’t Hooft’s formalism of Cellular Automata, [25, 26].

In Par.(4), by putting together the classical model for entanglement, the dynamics of “massless clocks” and the effect of polarizers, we will calculate the joint probability to observe a positive event in the two arms of the CHSH experiment. In Par.(4.1), we will read from the single Gen-CM expressions both the results of Std-CM, reproducing the result of ordinary LHVT, with no violation of CHSH’s inequality and the result of Cyclic CM, with violation of the CHSH inequality at a classical level, mirroring the predictions of standard QM. The result is interpreted in terms of Bell’s non-locality and Einstein’s locality in Par.(5), demonstrating that the perspectives of these two great scientists are not contradictory but rather complementary, ultimately leading to a reconciliation of their viewpoints.

This work, as the previous on the same topic, introduces a new paradigm of classical and quantum physics, necessitating the introduction of novel terminology. While we have striven to use clear and concise language, some terms may be unconventional or subject to varying interpretations. We encourage the reader to approach these terms with an open mind and refer back to the definitions provided within the text and previous work for clarification.

2 How Many Classical Mechanics?

The general definition of CM, here named for convenience Gen-CM, is the set of dynamics, derived from the PSA, requiring the action to be extremized both in the bulk and at the boundaries of a given time interval³. In this work, we distinguish two natural subclasses of Gen-CM, depending on the choice of the BCs.

³The PSA defines a complete Dirichlet problem: its variation yields the system’s EoMs from the bulk term (via the EL eqs), while the BCs, which select the specific solution, are determined by the boundary term. Importantly, consistently with the PSA, this boundary term can be minimized in different ways, leading to distinct subclasses of Gen-CM. See also footnote.(1)

We denote by $\psi(t)$ the general solution of motion in Gen-CM, prior to specifying the BCs that minimize the action at the boundary.

2.1 Standard Classical Mechanics

As is well known, the Std-CM, the formulation typically employed in classical physics, associated with the action

$$\mathcal{S} = \int_{t_i}^{t_f} dt L(q, \dot{q}), \quad (1)$$

where L is the Lagrangian, are the stationary solutions satisfying the null variation

$$\delta S = \int_{t_i}^{t_f} dt (\text{Euler Lagrange eqs}) \delta q + \left. \frac{\partial L}{\partial \dot{q}} \delta q \right|_{t_i}^{t_f} \equiv 0. \quad (2)$$

with vanishing variations at time endpoints t_i and t_f . In fact, the minimization of the action boundary is commonly obtained by imposing the familiar Std-BCs

$$\delta q(t_i) = \delta q(t_f) \equiv 0, \quad (3)$$

together with the Euler-Lagrange equations vanishing the 'bulk' term. This yields the Std-CM solution of local energy $E(t)$, denoted by $q(t)$ (highlighted in red in the text).

2.2 Cyclic Classical Mechanics

Cyclic-CM (leading to Std-QM) is the subclass of Gen-CM emerging when PBCs are imposed, rather than Std-BCs, enforcing intrinsic periodicity in time (and thus in space). These PBCs are fully compatible with the PSA, minimizing the action at the boundary just like Std-BCs of above. This compatibility ensures that, when properly implemented, PBCs preserve both causality and locality, as we will see.

Let us consider the following action where the time interval has been redefined as $t_i \rightarrow t$ and $t_f \rightarrow t + T(t)$:

$$\tilde{\mathcal{S}} = \int_t^{t+T(t)} dt L(\tilde{\psi}(t), \dot{\tilde{\psi}}(t)). \quad (4)$$

Again, we apply the PSA, requiring null variation

$$\delta \tilde{\mathcal{S}} = \int_t^{t+T(t)} dt (\text{Euler Lagrange eqs}) \delta \tilde{\psi} + \left. \frac{\partial L}{\partial \dot{\tilde{\psi}}} \delta \tilde{\psi} \right|_t^{t+T(t)} \equiv 0. \quad (5)$$

The action can be equally minimized by requiring PBCs at the time boundaries,

$$\tilde{\psi}(t) \equiv \tilde{\psi}(t + T(t)), \quad (6)$$

together with the EL eqs coming from the 'bulk' term. The solution of Cyclic-CM (highlighted in blue in the text) is denoted by $\tilde{\psi}(t)$ to distinguish it from the Gen-CM solution $\psi(t)$.

PBCs can be interpreted as a combination of two Dirichlet or Neumann conditions imposed at the temporal endpoints, similar to how such conditions are treated in string theory and Kaluza-Klein models. From a relativistic standpoint, temporal PBCs are as natural and consistent as spatial ones: due to Lorentz covariance, spatial periodicities in one frame transform into space-time periodicities in another, making temporal PBCs a direct consequence of relativistic covariance. Moreover, as we will show, adherence to the PSA implies that the time period $T(t)$ associated with the PBCs acquires a dynamical nature: it evolves covariantly along the Hamiltonian flow of the system⁴. This recurrence is determined locally by the system’s energy through the de Broglie relation (which takes the form of the Stokes’ theorem in Cyclic-CM), providing a fully local and causal description of Cyclic-CM.

As further discussed in Par.(5), the condition of IP expressed in eq.(6), though seemingly a source of manifest non-locality, is in fact a local and inherently classical nature, reconciling Bell’s request of non-locality with Einstein’s principle of locality.

Theorem (Statistical Equivalence Between Cyclic Classical Mechanics and Standard Quantum Mechanics). *For any Hamiltonian system, the statistical predictions of Cyclic-CM, derived via covariant constraint of intrinsic periodicity, are equivalent to those of Std-QM.*

We summarize results of paper [7] relevant for the interpretation of our results.

The solution of Cyclic-CM, $\tilde{\psi}(t)$, satisfies the standard EL eqs, but exhibits an explicitly wave-like nature due to the time recurrence imposed as a constraint — a feature that leads to the Schrödinger equation. Like a vibrating string, $\tilde{\psi}(t)$ is a Fourier superposition of harmonic modes selected by the PBCs, which in principle can be extended over all $t \in \mathbb{R}$.

The harmonic modes are determined by the local time recurrence $T(t)$ which, through the EoMs, imply a corresponding local spatial periodicity $\lambda(t)$. The following analysis provides a systematic method to implement the space-time recurrences intrinsic to any classical system, rooted in the Hamilton-Jacobi opto-mechanical analogy [17], where classical trajectories admit a dual wave description.

With particular reference to the spatial IP $\lambda(t)$, constituting the fundamental spatial recurrences of the harmonics set, the Cyclic-CM solution form a locally complete and orthogonal set, defining a Hilbert space structure. The system thus naturally admits a Hilbert space representation:

$$\tilde{\psi}(x) = \langle x | \tilde{\psi} \rangle \in \text{Cyclic Classical Mechanics} . \quad (7)$$

The state $|\tilde{\psi}\rangle$ is referred to as *ontic*, arising from a classical solution determined by the PSA. Ontologically, such states describe real, physically existing configurations — in contrast to the epistemic states of standard QM, which reflect statistical indeterminism. The ‘ontic’ states of Cyclic-CM must be understood as statistical representations of underlying ultra-fast cyclic dynamics (e.g., a “particle on a circle”). This

⁴Naively, one may note that any canonical transformation of the action variables induces a corresponding transformation of the boundary.

interpretation is consistent with deterministic approaches, such as 't Hooft's Cellular Automaton model [25, 26].

At time t , we adopt the following unitary normalization for the wave function of the Cycli-CM system over a single wavelength:

$$\langle \tilde{\psi} | \tilde{\psi} \rangle = \int_{-\lambda(t)/2}^{\lambda(t)/2} dx \tilde{\psi}^\dagger(x) \tilde{\psi}(x) = 1. \quad (8)$$

The reason for this normalization is in the fact that, in the case of Cyclic-CM, the 'ontic' state $|\psi\rangle$ can be imagined as describing at a statistical level a "particle moving very fast in a circle", [6–16]. It naturally defines an Hilbert space as also confirmed by 't Hooft's in terms of Cellular Automata, [25, 26]. In other words, similarly to a current, the probability to find the particle in every single cycle must be one. Thus $|\psi\rangle$ has the proper characteristic of a statistical distribution and Born's rule. That is, similar to Koopman-von Neumann mechanics, [27–29], we can associate a unitary probability to find the particle in the "circle".

We can generalize our analysis to any even-dimensional *symplectic manifold* equipped with a closed, non-degenerate 2-form Ω , known as the *symplectic form*, with $d\Omega = 0$. The symplectic structure also defines the Poisson bracket $\{\cdot, \cdot\}_{PB}$, where H is the Hamiltonian of the system — for instance that associated to the action $\tilde{S}$. The integral curves of X_H correspond to solutions of Hamilton's equations.

The condition of temporal IP eq. (6) can be introduced as a Dirac constraint

$$\oint_{\gamma_t} dt \frac{d}{dt} \tilde{\psi}(t) \equiv 0, \quad (9)$$

where we have introduced the notation $\int_t^{t+T(t)} dt = \oint_{\gamma_t} dt$ and γ_t denotes the closed time orbit determined by the local PBCs.

It leads to a constrained Hamiltonian of the form $\oint_{\gamma_t} dt \tilde{H} = \oint_{\gamma_t} dt (H - i\hbar \frac{d\psi}{dt})$, with $i\hbar$ acting as a multiplier. In the Hilbert space representation, the Hamiltonian can be promoted to an operator $\tilde{H} \rightarrow \hat{H}$. Dirac's consistency condition of the constraint of IP eq.(9) then becomes the ordinary *Schrödinger equation*:

$$\oint_{\gamma_t} dt \{ \cdot, \hat{H} - i\hbar \frac{d}{dt} \}_{PB} |\tilde{\psi}(t)\rangle \equiv 0. \quad (10)$$

The time-evolution operator is formally the same one of Std-CM: $U(t) = \exp[-\frac{i}{\hbar} \int_0^t \hat{H}(t) dt]$. The multiplier $i\hbar$ is determined by the Planck relation. The local energy spectrum of the Cyclic solution is

$$\oint_{\gamma_t} E_n(t') dt' = (n + \beta) \hbar, \quad (11)$$

with $n \in \mathbb{Z}$. We have included a Morse factor β which results from a twist factor in the PBCs — also allowed by the PSA. For instance, the temporal component ($\mu = 0$) for

the free case (persistent energy and time recurrence) and $\beta = 0$ reproduces the Planck spectrum of the electromagnetic radiation $E_n T = n h$ — or the energy spectrum of second quantization for $\beta = 1/2$ with vacuum energy $\frac{1}{2} \frac{h}{T}$.

Stokes' theorem guarantees that if eq.(11) holds at a given time t , then it holds in every other time, $\forall t \in \mathbb{R}$, *i.e.* $T(t)$ evolves covariantly with the Hamiltonian flow.

In this way, it is possible to establish a mathematical correspondence (mapping) between the Std-CM and the Cyclic-CM. Beside PBCs in time, we only assume Planck relation relating energy and time recurrences, by means of the Planck constant h . For this mapping to hold, the local recurrence $T(t)$ in the Cyclic-CM action must be determined by the local energy $E(t)$ of the corresponding Std-CM solution,

$$\oint_{\gamma_t} E(t') dt' = h, \quad \text{Mapping between Std-CM and the Cyclic-CM} . \quad (12)$$

— it reduces to the familiar Planck relation $ET = h$ in the free case.

Since $\oint_{\gamma_t}$ integrates over a period $T(t)$ that varies dynamically as a function of t , it is convenient to reformulate the problem using a constant reference period independent of time: $T_0 = \text{const}$, $\forall t \in \mathbb{R}$, represented by the integral $\oint_{\gamma_0}$. This leads to a covariant formulation where covariant derivatives, *e.g.* ∇_{X_H} , compensate for local modulations of the periodicity. For instance the state $|\psi\rangle$ is equally solution of both the two following expressions

$$\oint_{\gamma_t} dt \frac{d}{dt} |\tilde{\psi}\rangle = \oint_{\gamma_{t_0}} dt \nabla_{X_H} |\tilde{\psi}\rangle = 0 . \quad (13)$$

From this, the correct GQ pre-quantum of operator representations follow: $\tilde{H} \rightarrow \hat{H} = \tilde{H} - i\hbar \nabla_{X_H}$, $p \rightarrow \hat{p} = p - i\hbar \nabla_{X_p}$, $x \rightarrow \hat{x} = x - i\hbar \nabla_{X_x}$, being p the momentum. It is known from GQ that if operators take this form, their Poisson brackets reproduce Dirac's canonical quantization rule:

$$[\hat{x}, \hat{p}] = i\hbar \{x, p\}_{PB} = i\hbar \mathbb{I}, \quad (14)$$

and generally we have $[\hat{F}, \hat{G}] = i\hbar \{F, G\}_{PB}$ — we will further check this result in a naive independent way in Par.(3.1). These pre-quantum operators reproduce the standard operators of QM:

$$\hat{H}|\tilde{\psi}\rangle = i\hbar \frac{\partial}{\partial t} |\tilde{\psi}\rangle, \quad \hat{p}|\tilde{\psi}\rangle = -i\hbar \frac{\partial}{\partial x} |\tilde{\psi}\rangle, \quad \hat{x}|\tilde{\psi}\rangle = x|\tilde{\psi}\rangle .$$

This also means that Dirac's notation can be fully applied to Cyclic-QM as much as in Std-QM ⁵. See full detail in [7].

⁵This construction can be interpreted as a geometric deformation of the spacetime manifold: by imposing compactified temporal dimensions, the topology becomes non-trivial, inducing holonomies and necessitating connections on naturally occurring bundles such as the tangent bundle. As highlighted in [Referee Comment], this geometric structure naturally introduces the paraphernalia of quantum mechanics — including non-commutativity and operator algebra — via the constraint method generalized from Dirac's Hamiltonian treatment. Our results confirm this perspective, completing the quantization program via PBCs fully consistent with classical principles, including locality and causality.

This confirms the consistency of Cyclic-CM: its solution $\tilde{\psi}$ shares the same fundamental local space-time recurrences as standard de Broglie waves, constituting the fundamental mode of the harmonic set (for $\beta = 0$). But these common space-time recurrences are now imposed as constraints on the integrals of motion, providing a solution rich in harmonics. IP acts as the quantization condition of the classical system: a physical principle from which the full structure of Std-QM (Schrödinger equation, Hilbert space, commutators, operator algebra, etc.) can be derived rather than postulated.

Considering the induced IP on the spatial dimensions eq.(6) leads to a condition of closed space-time orbits γ_x in x , generalization of the Bohr-Sommerfeld quantization condition in space-time (being $\mu = 0$ the time component):

$$\oint_{\gamma_x} p_{n,\mu}(x) dx^\mu = (n + \beta)h. \quad (15)$$

The integral of the symplectic form Ω over the cyclic orbits must be an integer number in unit h (Weil integrability condition, fundamental requirement of the GQ program).

This detail is relevant to the discussion of Par.(5). In fact, according to Stokes' theorem, the space-time recurrences evolve along integral lines, in a contravariant way with respect to the energy-momentum, in order to remain on the same Hamiltonian vortex tube. The adherence to the PSA implies that the PBCs in time are dynamical and describe the correct modulation for instance of red-shifted clocks in a gravitational field or other types of interactions (including gauge interactions which have a natural geometrical origin in this formulation).

3 Set Up for Violation of Bell's Inequality in Classical Mechanics

To investigate the fundamental properties of Cyclic-CM that underlie the violation of the CHSH inequality [22, 23], we begin our analysis within the framework of Gen-CM. We want to describe 'photons' propagating in the arms of the CHSH experiment. From now on we will work in the assumption of free systems propagating at the speed of lights c , i.e. 'massless clock' with persistent and reciprocally proportional recurrences $\lambda = cT$ (we assume that both photons have the same wavelength λ).

Similarly to eq.(7) and eq.(8), it is convenient to represent such Gen-CM solutions $\psi(x)$ in a pseudo Dirac notation as Hilbert "ontic" states $|\psi\rangle$, such that $\psi(x) = [x|\psi]$:

$$|\psi\rangle \in \text{General Classical Mechanics}, \quad (16)$$

and related normalization,

$$[\psi|\psi] = \int_{-\lambda/2}^{\lambda/2} dx \psi^\dagger(x) \psi(x) = 1. \quad (17)$$

The pseudo Dirac notation $|\dots]$ (along with the omission of the tilde accent) indicates that we are in the framework of Gen-CM, where BCs are not explicitly imposed. Our strategy is to carry on calculation in this setting of Gen-CM, assuming Dirac's algebra. The physical predictions of the two subclasses will then be inferred by imposing the appropriate BCs at the end. In the subclass of Cyclic-CM, when PBCs are applied, these states acquire the full algebraic structure of quantum states [6–16], becoming strictly coherent with Dirac's bra-ket formalism of Std-QM denoted by $|\dots\rangle$ — though grounded in deterministic classical dynamics as suggested by the term 'ontic'. In the other subclass, by assuming Std-BCs, the same states become trivial: the Hilbert structure degenerates and commutators vanish, reflecting the absence of quantization and its trivially 'ontic' nature, so that we will be able to infer the results of Std-CM. We will further examine the consistency of this strategy in the following sections.

The selection of Cyclic-CM from the broader class of Gen-CM consists in imposing the condition of IP. For the purpose of our calculation, it will be sufficient to consider only the condition of spatial IP, without explicitly invoking the underlying fundamental temporal IP:

$$\left\{ \begin{array}{l} \psi(x) = \psi(x \bmod \lambda) \\ x = x \bmod \lambda \end{array} \right\} \equiv \left\{ \begin{array}{l} \text{Intrinsic} \\ \text{Periodicity.} \end{array} \right\}, \quad (18)$$

where the mapping between Cyclic-CM and Std-CM is given by identifying

3.1 The Classical Origin of Canonical Non-Commutativity

To demonstrate, again and independently, the connection of Cyclic-CM with Std-QM, a simple integration by parts will suffice for the rest of the paper; more comprehensive proofs are in [6–16] and Par.(2). Let ψ be a generic free solution of Gen-CM system, that is to say, in our analogy, a “massless clock” with characteristic persistent wavelength λ , and integrate by parts over a spatial recurrence λ (see Ehrenfest theorem, the following arguments can be applied also in terms of temporal IP):

$$\begin{aligned} [\psi|1|\psi]_{-\lambda/2}^{\lambda/2} &= [\psi|\partial_x x|\psi]_{-\lambda/2}^{\lambda/2} = \int_{-\lambda/2}^{\lambda/2} dx \psi^\dagger(x) (\partial_x x) \psi(x) \\ &= - \int_{-\lambda/2}^{\lambda/2} dx \psi^\dagger(x) [\overleftarrow{\partial}_x x + x \overrightarrow{\partial}_x] \psi(x) + \int_{-\lambda/2}^{\lambda/2} dx \partial_x [\psi^\dagger(x) x \psi(x)] \\ &= - \frac{i}{\hbar} [\psi|[x, p]|\psi]_{-\lambda/2}^{\lambda/2} + [\psi^\dagger(x) x \psi(x)]_{-\lambda/2}^{\lambda/2}. \end{aligned} \quad (19)$$

We have defined $p = -i\hbar\partial_x$ (corresponding, in our 'mapping' to the momentum of the dual classical solution $q(x)$). We also define a Gen-CM version of the annihilation and creation operators for the massless clock $\psi(x)$, such that: $x = \sqrt{\frac{1}{2\omega}}(a + a^\dagger)$ and $p = i\sqrt{\frac{\omega}{2}}(a^\dagger - a)$, hence $-\frac{i}{\hbar}[x, p] = [a^\dagger, a]$. We get,

$$[\psi|\psi]_{-\lambda/2}^{\lambda/2} = [\psi|[a^\dagger, a]|\psi]_{-\lambda/2}^{\lambda/2} + [\psi^\dagger(x) x \psi(x)]_{-\lambda/2}^{\lambda/2}. \quad (20)$$

Next we investigate separately the Cyclic-CM and the Std-CM resulting from this Gen-CM system.

3.1.1 Non-Commutativity of Cyclic Classical Mechanics (Equivalent to standard QM)

Consistently with the general proofs published in [6] for second quantization, in [7] for canonical quantization in all possible Hamiltonian mechanics, see Par.(2), and prior works also for the Feynman Path Integral [8–16], we illustrate how second quantization emerges naturally in CM as a consequence of imposing intrinsic spatial periodicity λ on the system, defined in eq.(18), rather than postulating non-commutativity of the operators as in standard quantum theory — the demonstration can be generalized to generic initial and final states [9, 10, 13–16].

From eq.(20), combined with eq.(8) and eq(18), we conclude that the characteristic non-commutativity observed in standard QM is inherent within the IP constraint of Cyclic-CM:

$$\left\{ \begin{array}{l} \psi(x) \stackrel{\text{PBCs}}{\equiv} \psi(x + \lambda) \\ x \equiv x + \text{mod } \lambda \end{array} \right\} \Leftrightarrow [\psi^\dagger(x)x\psi(x)]_{-\lambda/2}^{\lambda/2} \equiv 0 \Leftrightarrow a(x) \stackrel{\text{PBCs}}{\equiv} a(x + \lambda) \Leftrightarrow [a, a^\dagger] = 1 \quad (21)$$

In the forthcoming calculation, the condition of IP can be simply obtained by enforcing PBCs on the ladder operator, $a(x) \stackrel{\text{PBCs}}{\equiv} a(x + \lambda) \Leftrightarrow [a, a^\dagger] = 1$. This non-commutativity is proven in [6], constituting a further confirmation establishing the standard commutation algebra of ordinary QM in Cyclic-CM, without the need to postulate non-commutativity. Consequently, a vacuum state can be defined, and states can be created and annihilated using these operators, formally establishing quantum algebra in Cyclic-CM as we will further demonstrate shortly.

Summarizing, this simple argument illustrates the general result according to which standard QM is effectively reproduced by Cyclic-CM, a specific instance of Gen-CM where IP is imposed as constraint on the spatial coordinate x :

$$\left\{ \begin{array}{l} \text{Ordinary} \\ \text{Quantum} \\ \text{Mechanics} \end{array} \right\} \equiv \left\{ \begin{array}{ll} \text{General} & \text{Intrinsic} \\ \text{Classical} & \text{Periodicity} \\ \text{Mechanics} & \text{(PBCs)} \end{array} \right\} \equiv \left\{ \begin{array}{l} \text{Cyclic} \\ \text{Classic} \\ \text{Mechanics} \end{array} \right\}.$$

This concept will be used to demonstrate the existence of CM, specifically the Cyclic-CM, that violate Bell inequalities to the same extent as standard QM.

As we have seen in Par.(2), the equivalence between Cyclic-CM and standard QM is a general result, valid for all Hamiltonian systems and for arbitrary initial and final states — including interacting states. As detailed in [7], this equivalence encompasses the use of the Dirac formalism, Hilbert space, Schrödinger equation, the Feynman Path Integral, and other machinery of standard QM, within the framework of Cyclic-CM. While a more general treatment should be based on temporal recurrences [16], we will continue to focus for our calculation on spatial recurrence.

3.1.2 Commutativity of Standard Classical Mechanics

Std-CM represents standard classical solutions of the EoMs, characterized by stationary values at the boundaries rather than IP, *i.e.* Std-BCs. As we know, Std-CM is commutative $[a^\dagger, a] = 0$. In fact,

$$\left\{ \begin{array}{l} x \neq x + \text{mod } \lambda \\ \text{Standard BCs} \end{array} \right\} \Leftrightarrow [\psi^\dagger(x)x\psi(x)]_{-\lambda/2}^{\lambda/2} = [\psi|\psi]_{-\lambda/2}^{\lambda/2} = 1 \Leftrightarrow [a, a^\dagger] = 0. \quad (22)$$

Std-CM, including HVT, is obtained as a subclass of Gen-CM by imposing commutativity, *i.e.* Std-BCs, rather than IP

$$\left\{ \begin{array}{l} \text{General} \\ \text{Classical} \\ \text{Mechanics} \end{array} \quad \& \quad \begin{array}{l} \text{Standard} \\ \text{BCs} \\ \text{(commutativity)} \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \text{Standard} \\ \text{Classical} \\ \text{Mechanics} \end{array} \right.$$

and it satisfies Bell's inequalities as we will check below.

3.1.3 General Formalism for Cyclic and Standard Classical Mechanics

As already anticipated, our strategy is to perform calculations within the framework of Gen-CM, and then derive either the Cyclic-QM or the Std-QM results by imposing IP (canonical non-commutativity and thus Dirac's algebra) or Std-BCs (commutativity and trivial algebra) at the end of the calculations, respectively. This approach allows us to trace the origin of any violations of Bell's inequalities.

Since we are considering subclasses with either ordinary commutators of QM are implicit (Cyclic-CM) or vanishing commutators (Std-CM), we can proceed with our calculations assuming the standard quantum non-commutation rules together with Dirac's formalism for the Gen-CM calculations, and then extract the physical predictions by imposing either IP (non-commutativity) or Std-BCs (commutativity). In the former case, quantum algebra is naturally embedded; in the latter, it degenerates to classical algebra. We define the Gen-CM (pseudo) vacuum state and as $a|0\rangle = 0$, and construct (pseudo) states by acting on it with Gen-CM ladder operators (Gen-CM ladder operators as creation and annihilation operators). For example, $a^\dagger|0\rangle = |1\rangle$ with associated wave-function ψ_1 . The details are given in [7].

Based on the integration by parts described above, see eq.(20), we assume that the probability of observing an event within the framework of Gen-CM is

$$[1|a^\dagger a|1]_{-\lambda/2}^{\lambda/2} = [\psi_1^\dagger x \psi_1]_{-\lambda/2}^{\lambda/2} + [1|[a, a]|1]_{-\lambda/2}^{\lambda/2}, \quad (23)$$

where $\mathcal{N} = a^\dagger a$ is the number operator.

The statistical predictions of Cyclic-CM, which are identical to those of standard QM, are derived by imposing IP, corresponding to vanishing boundary term⁶. As

⁶Notice that in the case of Std-CM we have $\langle \psi_f | \psi_i \rangle_\lambda \rightarrow [\psi_f^\dagger(x)x\psi_i(x)]_{-\lambda/2}^{\lambda/2}$. This offers a hint to interpret Born rule in a classical way: the probability to observe a particle in the interval $[x_0 - \lambda/2, x_0 + \lambda/2]$ is given by the probability that the particle is passed in $x_0 - \lambda/2$ minus the probability that the particle

already said, this condition implicitly establishes the standard QM commutation relations in Cyclic-CM. For the Cyclic-CM the probability thus is given by the following relation

$$\text{Probability in Cyclic CM} : \left[\psi_1^\dagger x \psi_1 \right]_{-\lambda/2}^{\lambda/2} \equiv 0 \Leftrightarrow [1|a^\dagger a|1]_{-\lambda/2}^{\lambda/2} \stackrel{\text{PBCs}}{\equiv} [1|[a, a]|1]_{-\lambda/2}^{\lambda/2} = \langle 1|[a, a]|1 \rangle_{24} \quad (24)$$

where $\langle \rangle$ denotes that the result is formally equivalent to the corresponding quantity of standard QM, normalized over a wave-length, due to the implicit non-commutativity of the Cyclic-CM.

Conversely, the Std-CM result is obtained from the Gen-CM by simply assuming commutativity (Std-BCs), so that only the boundary term is left on the right side of eq.(23)

$$\text{Probability in Standard CM} : [1|[a, a]|1]_{-\lambda/2}^{\lambda/2} \equiv 0 \Leftrightarrow [1|a^\dagger a|1]_{-\lambda/2}^{\lambda/2} \stackrel{\text{Std-PBs}}{\equiv} \left[\psi_1^\dagger x \psi_1 \right]_{-\lambda/2}^{\lambda/2}. \quad (25)$$

In summary, the results of Std-CM (in red) are encoded in the boundary term of the integration by parts, eq.(23), while the results of Cyclic-CM (in blue) are encoded in bulk term. This calculation technique will be applied to describe the Cyclic-CM of our free “massless clocks”, mimicking the behavior of a quantum photon in the context of the CHSH experiment.

It is noteworthy that Std-CM can be interpreted as an effective, holographic description of Cyclic-CM, akin to the holographic description of extra-dimensional theories [20, 21], and Cyclic-CM is equivalent to standard QM in its predictions but fundamentally classical in nature. This is a possible way to show how, in ECT, our classical world (Std-CM) emerges from a more fundamental, yet still classical, level characterized by cyclic space-time dynamics (Cyclic-CM), realm beyond ordinary QM. Std-CM are the holographic projection of the Cyclic-CM, meant as the classical-relativistic physics beyond QM — thus Std-CM is the holographic projections of standard QM. As proven in [14] this reveals the real mathematical meaning of the Maldacena conjecture (AdS/CFT correspondence or Gauge/Gravity duality).

3.2 The Classical Origin of Entangled States

Next, we introduce a classical model that replicates the generation of entangled states, mirroring the two-stage $\Delta J = 0$ transitions of the CHSH experiment, [22, 23]. We formulate this model directly within the framework of Cyclic-CM, without need to consider the Std-CM case — the classical entangled state is created as it is, with no counterpart in Std-CM. The model consists in a specific instance of IP, analogous to case described previously, but now defined on a two-site lattice rather than in a continuum. Again, this discrete system, characterized by cyclic dynamics, exhibits a remarkable equivalence with a quantum system, as also noted by ’t Hooft [25, 26]. In essence, this is a two-site Periodic Cellular Automaton.

in passed in $x_0 + \lambda/2$. Otherwise, if intrinsic periodicity is imposed as constraint we get the ordinary Born rule of QM.

Entanglement in General CM

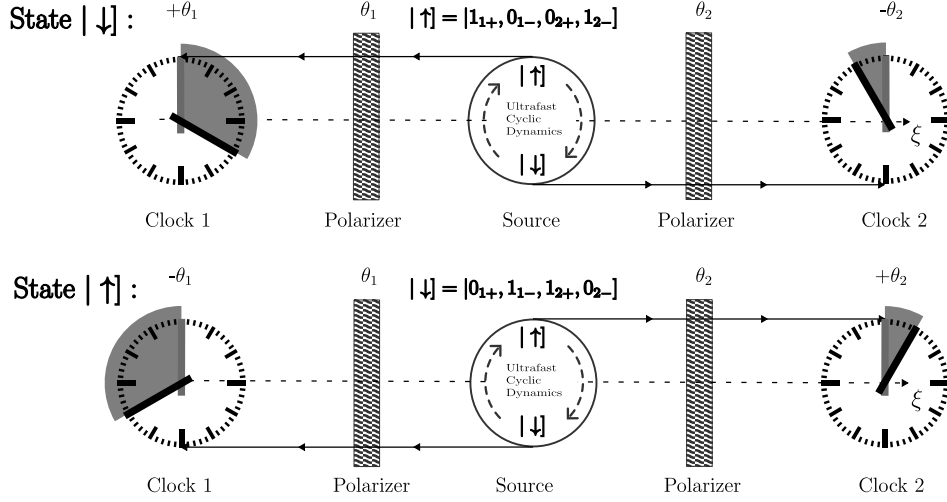


Fig. 2 This figure depicts the CHSH experiment within the framework of Cyclic-CM. The entangled source is modeled as intrinsically cyclic classical dynamics on a two-site lattice, oscillating between two classical “ontic” states, $|\uparrow\rangle$ and $|\downarrow\rangle$ at ultra-fast timescales, beyond the observer’s temporal resolution. The polarizer j , oriented at an angle θ_j , shifts the phase of the emitted signal’s cyclic dynamics (‘photons’) by an angle $\pm\theta_j$, depending on the state emitted.

In particular, by adopting ’t Hooft’s notation, we find that quantum entangled states can be reproduced at a classical level as a systems switching periodically, every ultra-small time interval T_S , among two possible classical “ontic” states: $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, see fig.(3.2)

Crucially, these intrinsically cyclic dynamics are assumed to occur on ultra-fast timescales T_S , faster than the Compton time of electrons⁷, far beyond the resolution of any modern resolution in time. This rapid switching, akin a coin tipping very fast, or a “tic-tac clock” mechanism operating at ultra-fast timescale, or a — two faces — ‘die’ rolling too fast to observe the underlying deterministic dynamics, effectively renders the system’s state observable only at a statistical level. Remarkably, the resulting deterministic statistical description of this IP system precisely mirrors the probabilistic, indeterministic outcomes predicted by QM.

Following the results of ECT and according to ’t Hooft’s formalism, [25, 26], the evolution of this IP system is governed by the following evolution operator:

$$\hat{U}_{op}(T_S) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (26)$$

⁷The emission of photons from an atom — as much as all detection systems — is governed by electrodynamics, whose temporal dynamics are inevitably determined by the internal cyclic dynamics of the electrons and thus faster or equal to the electron Compton time $T_C = h/m_e c^2 \sim 10^{-21} \text{s}$, being m_e the mass of the electron. This sets a fundamental threshold below which the ultra-fast dynamics predicted by Cyclic-CM cannot be directly resolved. At timescales above this limit, the Cyclic-CM can only be accessed through statistical descriptions.

Its diagonalization implies the following eigenvectors

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle) \quad , \quad |\Psi'\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle). \quad (27)$$

In this model there is the possibility to introduce a Hamiltonian operator, $\mathcal{H}_{op}$. Remarkably, the system's evolution is then described by the ordinary Schrödinger equation and satisfies the desired correspondences with ordinary QM, including implicit non-commutativity induced by the IP, while still maintaining its inherently classical nature, [6–16], as also confirmed by 't Hooft work [25, 26].

To establish a classical description of a source of singlet-state entanglement, we must also introduce a selection mechanism that isolates only the state $|\Psi\rangle$ from the two eigenvectors of eqs.(27).

4 Violation of Bell's Inequality in Classical Mechanics

We define each of the “ontic” states $|\uparrow\rangle$ and $|\downarrow\rangle$ of the relevant eigenvector $|\Psi\rangle$ in terms of two Gen-CM free “massless clocks” labeled 1 and 2 of the kind described in Par.(3.1), traveling in opposite directions along the z -axis at the speed of light. To model the $\Delta J = 0$ source associated to $|\Psi\rangle$, we assume that the “clocks” have transverse oscillations in orthogonal directions. Specifically, the two “ontic” states are defined as $|\uparrow\rangle = |1_{1x}, 0_{1y}, 0_{2x}, 1_{2y}\rangle$ and $|\downarrow\rangle = |0_{1x}, 1_{1y}, 1_{2x}, 0_{2y}\rangle$, where $|1_{jx}\rangle = a_{jx}^\dagger|0\rangle$ represents “clock” j oscillating in the x -direction, as allowed by the pseudo Dirac algebra of the single components of the two Gen-CM ‘clocks’, see Par.(2) and Par.(3):

$$|\Psi\rangle \equiv \frac{1}{\sqrt{2}}(|1_{1x}, 0_{1y}, 0_{2x}, 1_{2y}\rangle - |0_{1x}, 1_{1y}, 1_{2x}, 0_{2y}\rangle). \quad (28)$$

The effect of a polarizer oriented at an angle θ_j on the “clock” j is to rotate the direction of transverse oscillation by an angle θ_j from the x -axis, [22, 23]. Within the framework of Gen-CM, this effect can be described as follows:

$$a_j = a_{jx} \cos \theta_j + a_{jy} \sin \theta_j \quad (29)$$

where a_j and a_{jx} or a_{jy} are the Std-CM dynamical variable behind and before the polarizer.

4.0.1 The Phase Shift on the Condition of Intrinsic Periodicity induced by the Polarizers

We find that the effect of a polarizer on a ‘massless clock’ is to induce a positive or negative phase shift in its oscillation. This phase shift plays a crucial role in our analysis, as it directly influences the specific conditions of IP required to obtain the correspondence with standard QM and in turn to achieve violations of the CHSH inequality. As a rule of thumb, we must remember that the condition of IP is in fact determined by the recurrence of the wave-function, [6–16].

For simplicity, we assume that both 'massless clocks' exhibit the same spatial recurrence $\lambda = \pi$ when propagating away before and after the polarizers. Consequently, the spatial recurrence of the combined state $|\Psi\rangle$ is $\pi/2$, resulting from the product of these two π -periodic phenomena. The $\pi/2$ intervals is also the reference length for the normalization of the wave-function $|\Psi\rangle$ of the system.

For this scope it is convenient to adopt the following substitution in place of eq.(29),

$$a_j = \frac{1}{\sqrt{2}}(a_{j+}e^{i\theta_j} + a_{j-}e^{-i\theta_j}), \quad (30)$$

so that Gen-CM "entangled state" reads

$$|\Psi\rangle \equiv \frac{1}{\sqrt{2}}(|1_{1+}, 0_{1-}, 0_{2+}, 1_{2-}\rangle - |0_{1+}, 1_{1-}, 1_{2+}, 0_{2-}\rangle). \quad (31)$$

where $a_{j+} = \frac{1}{\sqrt{2}}(a_{jx} - ia_{jy})$ and $a_{j-} = \frac{1}{\sqrt{2}}(a_{jx} + ia_{jy})$ and, *e.g.*, $|1_{j\pm}\rangle = a_{j\pm}^\dagger|0\rangle$ is the related pseudo-Dirac algebra.

As the "clock" j interacts with a polarizer oriented at an angle θ_j , it undergoes a sudden phase shift: $\psi_{j\pm}(z) \rightarrow \psi_{j\pm}(z \mp \theta_j)$. This phase shift necessitates a corresponding adjustment of the boundaries of the integration by parts in eq.(19), which must be displaced accordingly. Consequently, the condition of IP required to establish Cyclic-CM involves a spatial coordinate shift $z \rightarrow z \mp \theta_j$ for clock j with respect to the free case eq.(18). In particular, we have the shift on the parameter of the dynamical variable $a_{j\pm}(z) \rightarrow a_{j\pm}(z \mp \theta_j)$ as the "clock" passes through the polarizer, see also [30–33].

Thus, the interaction with the polarizer modifies the IP condition with respect to the free case. The boundary of the integration by part eq.(19) are shifted by an amount $\pm\theta_j$, corresponding to the phase shift experienced by "clock". This shift, however, has no significant impact on states of the form $|1_{j\pm}\rangle a_{j\pm}^\dagger|1_{j\pm}\rangle$, as the displacements in the two boundaries are in the same directions. Thus their effect cancel each other out. In this case the shift will not be reported in the calculations. In contrast, for states like $|1_{j\pm}\rangle a_{j\pm}^\dagger|1_{j\mp}\rangle$, the shifts occur in opposite directions, contributing the final result.

As demonstrated in the previous sections and rigorously proven in [6], all ladder operators within the framework of Cyclic-CM satisfy the same algebraic relations as their counterparts in standard QM, which in this case are:

$$\text{Cyclic-CM: } [\hat{a}_{j\pm}, \hat{a}_{i\pm}^\dagger] = \delta_{ij} \quad , \quad [\hat{a}_{j\pm}, \hat{a}_{i\mp}^\dagger] = 0, \quad (32)$$

while they vanish in the case of Std-CM. The hat symbol represents the fact that the operators do not commute in the framework Cyclic-CM, exactly mimicking the standard QM operators.

4.1 Joint Probability According to General Classical Mechanics

We can now formulate the joint probability within the framework of Gen-CM for both signals to pass their respective polarizers. This probability is assumed to be given by,

[22, 23],

$$P_{\oplus, \oplus}^{Gen-CM}(\theta_1, \theta_2) = [\Psi | a_1^\dagger a_2^\dagger a_2 a_1 | \Psi]. \quad (33)$$

The notation is further simplified by introducing

$$A_\uparrow = a_{1+} a_{2-} \quad , \quad A_\downarrow = a_{1-} a_{2+}. \quad (34)$$

By taking into account the displacement of the boundaries induced by the polarizes the joint probability of Gen-CM, eq.(33), is

$$P_{\oplus, \oplus}^{Gen-CM}(\theta_1, \theta_2) = \frac{1}{8} \left\{ [\uparrow | A_\uparrow^\dagger A_\uparrow | \uparrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - [\uparrow | A_\uparrow^\dagger e^{i2\Delta\theta} A_\downarrow | \downarrow]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} \right. \\ \left. - [\downarrow | A_\downarrow^\dagger e^{-i2\Delta\theta} A_\uparrow | \uparrow]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} + [\downarrow | A_\downarrow^\dagger A_\downarrow | \downarrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \right\}. \quad (35)$$

The final result must be invariant under the exchange of polarizer orientations, depending solely on the absolute value of their angle difference: $|\Delta\theta| = |\theta_2 - \theta_1|$. The boundary of the mixed terms is set to $\pm \frac{\pi}{4} - |\Delta\theta|$ to ensure that the calculated expectation values remain within the physically meaningful range of 0 to 1 (the inversion of the order between the initial and final points must be accompanied by a corresponding sign inversion in the integral).

Explicit calculation of the individual terms is achieved through integration by parts, similarly to eq.(23). In the evaluation we also recursively apply the commutation relations, keeping in mind that they are either the standard one, eq.(32), for Cyclic-CM or they are systematically vanishing for the Std-CM. See Appendix.(A) for the explicit calculations.

The first and fourth terms are independent of $\Delta\theta$ as their associated boundary terms are displaced in the same direction. The second and third terms exhibit a dependence on $|\Delta\theta|$, arising from the fact that their respective boundary terms are displaced in opposite directions:

$$[\uparrow | A_\uparrow^\dagger A_\uparrow | \uparrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = [\Psi_\uparrow^\dagger z \Psi_\uparrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} + [\uparrow | [A_\uparrow, A_\uparrow^\dagger] | \uparrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}, \\ [\downarrow | A_\downarrow^\dagger A_\downarrow | \downarrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = [\Psi_\downarrow^\dagger z \Psi_\downarrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} + [\downarrow | [A_\downarrow, A_\downarrow^\dagger] | \downarrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}, \\ [\uparrow | A_\uparrow^\dagger e^{i2\Delta\theta} A_\downarrow | \downarrow]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} = [\Psi_\uparrow^\dagger e^{i2\Delta\theta} z \Psi_\downarrow]_{-\frac{\pi}{2}-|\Delta\theta|}^{\frac{\pi}{2}-|\Delta\theta|} \\ + [\uparrow | e^{i2\Delta\theta} [A_\uparrow, A_\downarrow^\dagger] | \downarrow]_{-\frac{\pi}{2}-|\Delta\theta|}^{\frac{\pi}{2}-|\Delta\theta|}, \\ [\downarrow | A_\downarrow^\dagger e^{-i2\Delta\theta} A_\uparrow | \uparrow]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} = [\Psi_\downarrow^\dagger e^{-i2\Delta\theta} z \Psi_\uparrow]_{-\frac{\pi}{2}-|\Delta\theta|}^{\frac{\pi}{2}-|\Delta\theta|} \\ + [\downarrow | e^{-i2\Delta\theta} [A_\uparrow, A_\downarrow^\dagger] | \uparrow]_{-\frac{\pi}{2}-|\Delta\theta|}^{\frac{\pi}{2}-|\Delta\theta|}. \quad (36)$$

Next we will extract separately the Cyclic-CM and Std-CM results by imposing IP (vanishing the boundary terms), yielding results that precisely match those of QM, or by assuming commutativity to obtain the results of Std-CMs, identical to those of LHVT.

4.1.1 Joint Probability for Cyclic Classic Mechanics: Violation of Bell's Inequality

The violation of the CHSH inequality is extracted from the Gen-CM joint probability eq.(33) by simply imposing IP, which means vanishing boundary terms. These are the Cyclic-CM where the commutation relations eq.(32) are implicitly satisfied at a classical level (without postulating them). Thus,

$$\begin{aligned} [\Psi_{\uparrow}^{\dagger} z \Psi_{\uparrow}]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = 0 &\Leftrightarrow [\uparrow |[\hat{A}_{\uparrow}, \hat{A}_{\uparrow}^{\dagger}]| \uparrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \stackrel{\text{PBCs}}{\equiv} \langle \uparrow |[\hat{A}_{\uparrow}, \hat{A}_{\uparrow}^{\dagger}]| \uparrow \rangle = 1, \\ [\Psi_{\downarrow}^{\dagger} z \Psi_{\downarrow}]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = 0 &\Leftrightarrow [\downarrow |[\hat{A}_{\downarrow}, \hat{A}_{\downarrow}^{\dagger}]| \downarrow]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \stackrel{\text{PBCs}}{\equiv} \langle \downarrow |[\hat{A}_{\downarrow}, \hat{A}_{\downarrow}^{\dagger}]| \downarrow \rangle = 1. \end{aligned} \quad (37)$$

Again $\langle \rangle$ indicates that Cyclic-CM reproduces the same result of ordinary QM.

The vanishing boundary terms, implying the commutators eq.(32), in turn also yields

$$\begin{cases} [\Psi_{\uparrow}^{\dagger} e^{i2\Delta\theta} z \Psi_{\downarrow}]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} = 0 \\ [\Psi_{\downarrow}^{\dagger} e^{-i2\Delta\theta} z \Psi_{\uparrow}]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} = 0 \end{cases} \quad (38)$$

and thus⁸

$$\begin{cases} [\uparrow |e^{i2\Delta\theta} [A_{\uparrow}, A_{\downarrow}^{\dagger}]| \downarrow]_{-\frac{\pi}{2}-|\Delta\theta|}^{\frac{\pi}{2}-|\Delta\theta|} \stackrel{\text{PBCs}}{\equiv} \langle \uparrow |e^{i2\Delta\theta} [\hat{A}_{\uparrow}, \hat{A}_{\downarrow}^{\dagger}]| \downarrow \rangle = e^{i2\Delta\theta}. \\ [\downarrow |e^{-i2\Delta\theta} [A_{\uparrow}, A_{\downarrow}^{\dagger}]| \uparrow]_{-\frac{\pi}{2}-|\Delta\theta|}^{\frac{\pi}{2}-|\Delta\theta|} \stackrel{\text{PBCs}}{\equiv} \langle \downarrow |e^{-i2\Delta\theta} [\hat{A}_{\downarrow}, \hat{A}_{\uparrow}^{\dagger}]| \uparrow \rangle = e^{-i2\Delta\theta} \end{cases} \quad (39)$$

These results are identical to those obtained in standard QM, represented by the quantity $\langle \rangle$, despite being derived entirely within the framework of Gen-CM. This agreement is achieved by imposing IP on the system, in full adherence to the PLA. Thus we have local CM. The joint probability for Cyclic-CM is thus:

$$\text{Cyclic CM: } P_{\oplus, \oplus}^{\text{Cyclic-CM}}(\theta_1, \theta_2) = \frac{1}{2} \sin^2(\theta_1 - \theta_2). \quad (40)$$

As well-known, [22, 23], a direct consequence of this result is the violation of Bell's inequalities. Remarkably, in contrast to conventional belief, we have just demonstrated

⁸Here the corresponding terms of eq.(37) must be properly rescaled by normalization factors in order to obtain the QM result which is normalized to a single wave-length π

the existence of a class of CM, named here Cyclic-CM, which exhibits Bell's inequality violations to the same extent as observed in QM.

Several points require particular care in the above derivation. The phase factors $e^{\pm 2\Delta\theta}$ in the boundary terms adjust precisely the recurrence of wave-functions so that they have identical values at the shifted boundaries. At the same time the presence of a polarizer implies the modification of the IP condition on the coordinate z in order to obtain vanishing boundary terms. As already said, the IP acquires a positive or negative shift $\Delta\theta$ as the signal passes through the polarizers. These shifts cancel in the homogeneous terms and sum in the non-homogeneous terms. IP means that z takes the same values at the two boundaries. Finally, it's important to note that the normalization of the probabilities must be performed over an unperturbed period of $\pi/2$, as the wave function $|\psi\rangle$ is normalized with respect to this period.

4.1.2 Joint Probability for Standard Classical Mechanics: Bell's Inequality

The result of Std-CM, characterized by Std-BCs, emerges from the Gen-CM by simply assuming commutativity among all the ladder operators. The only nonzero contribution in the right hand side of eq.(48) comes from the boundary terms, which, unlike in the case of Intrinsic Periodicity (IP), now do not vanish:

$$\begin{aligned} \left[\uparrow |A_{\uparrow, \Delta\theta}^\dagger A_{\uparrow, \Delta\theta}| \uparrow \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} &\stackrel{\text{Std-BCs}}{=} \left[\Psi_{\uparrow}^\dagger z \Psi_{\uparrow} \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = 1, \\ \left[\downarrow |A_{\downarrow, \Delta\theta}^\dagger A_{\downarrow, \Delta\theta}| \downarrow \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} &\stackrel{\text{Std-BCs}}{=} \left[\Psi_{\downarrow}^\dagger z \Psi_{\downarrow} \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = 1, \end{aligned} \quad (41)$$

and

$$\begin{aligned} \left[\uparrow |A_{\uparrow}^\dagger e^{i2\Delta\theta} A_{\downarrow}| \downarrow \right]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} &\stackrel{\text{Std-BCs}}{=} \left[\Psi_{\uparrow}^\dagger e^{i2\Delta\theta} z \Psi_{\downarrow} \right]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} = \left| 1 - \frac{2|\Delta\theta|}{\pi} \right| \\ \left[\downarrow |A_{\downarrow}^\dagger e^{-i2\Delta\theta} A_{\uparrow}| \uparrow \right]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} &\stackrel{\text{Std-BCs}}{=} \left[\Psi_{\downarrow}^\dagger e^{-i2\Delta\theta} z \Psi_{\uparrow} \right]_{-\frac{\pi}{4}-|\Delta\theta|}^{\frac{\pi}{4}-|\Delta\theta|} = \left| 1 - \frac{2|\Delta\theta|}{\pi} \right| \end{aligned} \quad (42)$$

The contributions of the phase shifts $e^{\pm 2\Delta\theta}$ play again a crucial role in the evaluation of the two boundary terms. These phase factors ensure that the wave-functions assume identical values at the shifted boundaries as in the previous case. However, in this scenario, the condition of IP is not applicable for the coordinate z , leading to different values at the the two boundaries of each expression, and thus to the results above.

In conclusion, the result obtained within the framework of Std-CM is the prediction of LHV theories:

$$\text{Standard CM: } P_{\oplus, \oplus}^{\text{Std-CM}}(\theta_1, \theta_2) = \frac{1}{2} \left(1 - \left| 1 - \frac{2|\Delta\theta|}{\pi} \right| \right). \quad (43)$$

To ensure proper probability normalization within the framework of Std-CM, the probability amplitudes must be multiplied by a factor of 2. This normalization factor arises from the difference in recurrence periods between the two-photon entangled state $|\Psi\rangle$ and the 'single-photon' state within $|\psi_j\rangle$. As already said, the 'two-photon' state exhibits a recurrence period $\pi/2$ while the 'single-photon' state has a recurrence period of π . Cyclic-CM, governed by the condition of IP, inherently enforces probability normalization with respect to the correct recurrence period of $\pi/2$ in the Cyclic-CM. The condition of IP in fact says that the unitary normalization must be on the elementary recurrence, analogous to the concept of a 'particle on a circle' discussed in Par.(3.2). The probability of observing two 'photons' within the Std-CM framework is performed with respect to the original period π , being not affected by the condition of IP, leading to a double-counting of events.

5 Conciliating Bell's Non-Locality and Einstein's Locality

According to ECT, [6–16] whose foundational aspects are reported in Par.(2), in Cyclic-CM, when interactions are present, the space-time IP becomes a dynamical variable: a contravariant four-vector modulated along the system's Hamiltonian flow. According to Stokes' theorem, see eq.(15), IP couples to the covariant four-momentum to yield the relativistic invariant h .

Importantly, *the local transformation of the IP, which must be understood as a dynamical variable, prevents that the Lorentz invariance is broken in ECT, as a consequence of its adherence to the classical PSA.* The time recurrence $T(x)$ and spatial recurrence $\lambda(x)$ are locally determined by the local energy-momentum via the de Broglie relation, which, imposed as constraint, takes the form a generalized Bohr-Sommerfeld quantization or Stoke's theorem, eq.(15). It represents quantization condition equivalent to the PBCs eq.(6) — condition of closed spacetime orbits. Only interactions within the light cone can influence these modulations, ensuring that causality and locality are respected. Thus, the condition of IP is a local and dynamical principle consistent with relativistic classical physics — and with gravitational and gauge interactions [7, 15].

Despite its classical foundation, Cyclic-CM exhibits a unique form of correlation that resembles Bell's notion of non-locality when considering systems that are supposed to have remained entirely isolated, even at space-like separated points. This non-locality, while permissible within the classical PSA, is contingent upon the hypotheses of absence of interactions. In the absolute absence of interactions, a physical quantity exhibiting persistent IP in time and space will have identical values at all points separated by integer multiples of its spatial and temporal periods $x + n\lambda$ or $t + nT$. This implies an inherent non-locality, as knowledge of the quantity's value at one point instantaneously reveals its value at distant space-like separated points, assuming that any interaction — variation of energy — is occurred (persistent IP) to alter the recurrences. This type of "non-locality" underpins the violation of Bell's inequality observed in this work and could be interpreted as a "spooky action at a distance" if the dynamical nature of IP is not considered.

On the other hand, if interaction occurs at space-like distance, in that space-time point the recurrences will be alternated with respect to the persistent (free) case, and the value of the physical quantity at that distant point will no longer be predictable. The apparent incompatibility between the non-local nature of Std-QM and local nature of classical mechanics is resolved by recognizing the crucial role of interactions in modulating the apparently “non-local” condition of IP.

Cyclic-CM resolves the apparent conflict between Einstein’s demand for locality and Bell’s demonstration of non-locality of nature. Both perspectives are valid within Cyclic-CM, and this long-standing debate is finally reconciled. QM emerges from fundamentally classical, deterministic physics rooted in the ultra-fast cyclic dynamics associated to every massive elementary particle. According to ECT every elementary particle of mass m can be seen as an elementary clock, ticking at a rate of $T_C = h/mc^2$ (see also de Broglie internal clock). These rates are incredibly fast. As an example, an electron, a light particle, exhibits an IP of the order of the 10^{-21} sec — see footnote.(7). This is an IP on the proper time of the particle, is the intrinsic recurrence of the rest particle. The case of photons is particularly intriguing. As expected from relativistic principles, the internal clock of a photon appears frozen ($T_C = \infty$). Observing its rest IP would require traveling at the speed of light, leading to infinite time dilation for the observer. After all this dynamical nature of time recurrences is also implicit in GR, which can be consistently described in terms of clock modulations and ruler expansions induced by the gravitational field, [34].

Beyond reproducing QM, ECT extends the clock-based interpretation of General Relativity to all interactions, including gauge interaction. As fully proven in [15] and generally shown by eq.(13), gauge interactions can be derived from classical Principle of General Invariance, without postulating gauge symmetry — in realization of original Weyl’s proposal. This leads to a unified description where all interactions (including gauge interactions) modulate the internal clocks of particles — keeping them perfectly synchronized since the beginning of time — offering conceptual links to superdeterminism [25, 26].

Finally, we must note that Cyclic-CM doesn’t involve any hidden variables, being based exclusively on the geometry of the spacetime coordinates and PBCs determined by the Planck constant which relates the spacetime recurrences with the four-momentum of the particles. This point is crucial because Bell’s theorem, which relies on the assumption of local hidden variables, cannot be therefore invoked to rule out Cyclic-CM as a viable classical mechanics at the base of QM.

6 Conclusions

In past works, we have developed a classical theory, namely Elementary Space-Time Cycles Theory (ECT), [6–16], strongly indicating a possible derivation of both standard QM and relativistic mechanics in a unified way directly from the deterministic classical Principle of Stationary Action (PSA) and without involving hidden variables of any sort — moreover, within ECT, both gauge theories and gravity emerge from the General Principle of Invariance, see [15], and subsequent papers.

The formal analysis of this paper indicates that Bell’s inequality is violated at a classical level within the subclass of Classical Mechanics, named here Cyclic Classical Mechanics (Cyclic-CM), framework inherent to ECT, to the same extent as in standard QM. Conversely, Bell’s inequality is satisfied in the other subclass which is the Standard Classical Mechanics (Std-CM) — the common mechanics used to study classical systems — including LHVT. By developing a novel calculation technique, we show that Std-CM emerges as a “holographic projection” of the more fundamental Cyclic-CM, further establishing ECT as the perfect candidate for a classical theory that encompasses and extends beyond QM.

The crucial element for a complete classical description of QM, in the spirit of EPR, is the assumption of intrinsically cyclic dynamics at ultra-fast scales for the elementary particles, whose recurrence periods must be understood as dynamical function of the local energy and therefore preserving causality and locality. This Intrinsic Periodicity (IP) naturally gives rise to the non-commutativity of conjugate variables observed in standard QM, without requiring its postulation as an axiom, as proven in the most general way in [7] for every possible Symplectic system (including systems with infinite degrees of freedom such as in field theory and in Riemann geometries such as in GR).

The physical principle IP seems to resolve the apparent conflict between Einstein’s and Bell’s views on QM. Like a relativistic clock ticking periodically, every elementary particle exhibits a recurring behavior predictable in time, even beyond the light cone, if total absence of interactions is assumed. This inherent predictability embodies Bell’s non-locality, which, while compatible with General Relativity, interactions, such as those modeled by polarizers in this paper, causally alter these recurrences, reflecting Einstein’s locality. Therefore, Cyclic-CM is a valid candidate for the conciliation of both: Bell’s non-locality (predictable recurrence of elementary particle dynamics) and Einstein’s locality (local, causal modulation of cyclic dynamics meant of dynamical character similar to relativistic clocks in GR). An important confirmation of our theoretical analysis would be the implementation of a classical algorithm, based on the Cyclic-CM described in our works, that exhibits a violation of Bell’s inequality, without invoking hidden variables.

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7 Declarations

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A Explicit Calculation of the Joint Probability

We have the following useful identities for commutators among generic operators A , B , C and D

$$[AB, CD] = A[B, C]D + C[A, D]B + \frac{1}{2}[A, C](BD + DB) + \frac{1}{2}[B, D](CA + AC) \quad (44)$$

Its recursive application on the Gen-CM the left side of the expectation values eq.(48) yields

$$\begin{aligned} [\uparrow |a_{\uparrow}^{\dagger} a_{\uparrow}| \uparrow] &= \frac{1}{2}[0|a_{1+}a_{1+}^{\dagger} + a_{2-}a_{2-}^{\dagger}|0] \\ &= \frac{1}{2}[1_{1+}, 0_{1-}, 0_{2+}, 0_{2-}|a_{1+}^{\dagger}a_{1+}|1_{1+}, 0_{1-}, 0_{2+}, 0_{2-}] \\ &\quad + \frac{1}{2}[0_{1+}, 0_{1-}, 0_{2+}, 1_{2-}|a_{2-}^{\dagger}a_{2-}|0_{1+}, 0_{1-}, 0_{2+}, 1_{2-}] \\ [\downarrow |a_{\downarrow}^{\dagger} a_{\downarrow}| \downarrow] &= \frac{1}{2}[0|a_{1-}a_{1-}^{\dagger} + a_{2+}a_{2+}^{\dagger}|0] \\ &= \frac{1}{2}[0_{1+}, 1_{1-}, 0_{2+}, 0_{2-}|a_{1-}^{\dagger}a_{1-}|0_{1+}, 1_{1-}, 0_{2+}, 0_{2-}] \\ &\quad + \frac{1}{2}[0_{1+}, 0_{1-}, 1_{2+}, 0_{2-}|a_{2+}^{\dagger}a_{2+}|0_{1+}, 0_{1-}, 1_{2+}, 0_{2-}] \\ [\uparrow |a_{\uparrow}^{\dagger} a_{\downarrow}| \downarrow] &= \frac{1}{4}[0|a_{1+}a_{1+}^{\dagger} + a_{1-}a_{1-}^{\dagger} + a_{2+}a_{2+}^{\dagger} + a_{2-}a_{2-}^{\dagger}|0] \\ &= \frac{1}{4}[1_{1+}, 0_{1-}, 0_{2+}, 0_{2-}|a_{1+}^{\dagger}a_{1+}|1_{1+}, 0_{1-}, 0_{2+}, 0_{2-}] \\ &\quad + \frac{1}{4}[0_{1+}, 1_{1-}, 0_{2+}, 0_{2-}|a_{1-}^{\dagger}a_{1-}|0_{1+}, 1_{1-}, 0_{2+}, 0_{2-}] \\ &\quad + \frac{1}{4}[0_{1+}, 0_{1-}, 1_{2+}, 0_{2-}|a_{2+}^{\dagger}a_{2+}|0_{1+}, 0_{1-}, 1_{2+}, 0_{2-}] \\ &\quad + \frac{1}{4}[0_{1+}, 0_{1-}, 1_{2+}, 0_{2-}|a_{2-}^{\dagger}a_{2-}|0_{1+}, 0_{1-}, 1_{2+}, 0_{2-}] = [\downarrow |a_{\downarrow}^{\dagger} a_{\uparrow}| \uparrow] \quad (45) \end{aligned}$$

In this form it is possible to perform the integration by parts analysis, similarly to eq.(19), isolating the boundary terms containing the result of Std-CM and the bulk terms containing the results of Cyclic-CM. For brevity, the boundary values are not explicitly stated. Thus:

$$\begin{aligned} [\uparrow |a_{\uparrow}^{\dagger} a_{\uparrow}| \uparrow] &= \frac{1}{2}[\Psi_{1+}^{\dagger} z \Psi_{1+}] + \frac{1}{2} \int dz \Psi_{1+}^{\dagger} [a_{1+}, a_{1+}^{\dagger}] \Psi_{1+} \\ &\quad + \frac{1}{2}[\Psi_{2-}^{\dagger} z \Psi_{2-}] + \frac{1}{2} \int dz \Psi_{2-}^{\dagger} [a_{2-}, a_{2-}^{\dagger}] \Psi_{2-} \\ [\downarrow |a_{\downarrow}^{\dagger} a_{\downarrow}| \downarrow] &= \frac{1}{2}[\Psi_{1+}^{\dagger} z \Psi_{1+}] + \frac{1}{2} \int dz \Psi_{1+}^{\dagger} [a_{1+}, a_{1+}^{\dagger}] \Psi_{1+} \end{aligned}$$

$$+ \frac{1}{2}[\Psi_{2-}^\dagger z \Psi_{2-}] + \frac{1}{2} \int dz \Psi_{2-}^\dagger [a_{2-}, a_{2-}^\dagger] \Psi_{2-} . \quad (46)$$

where z describes the coordinate along the propagation of the “photons”.

Similarly we evaluate the second and third terms:

$$\begin{aligned} [\uparrow | a_\uparrow^\dagger e^{i2\Delta\theta} a_\downarrow | \downarrow] &= \frac{1}{4}[\Psi_{1+}^\dagger e^{i2\Delta\theta} z \Psi_{1+}] + \frac{1}{4} \int dz e^{i2\Delta\theta} \Psi_{1+}^\dagger [a_{1+}, a_{1+}^\dagger] \Psi_{1+} \\ &+ \frac{1}{4}[\Psi_{1-}^\dagger e^{i2\Delta\theta} z \Psi_{1-}] + \frac{1}{4} \int dz e^{i2\Delta\theta} \Psi_{1-}^\dagger [a_{1-}, a_{1-}^\dagger] \Psi_{1-} \\ &+ \frac{1}{4}[\Psi_{2+}^\dagger e^{i2\Delta\theta} z \Psi_{2+}] + \frac{1}{4} \int dz e^{i2\Delta\theta} \Psi_{2+}^\dagger [a_{2+}, a_{2+}^\dagger] \Psi_{2+} \\ &+ \frac{1}{4}[\Psi_{2-}^\dagger e^{i2\Delta\theta} z \Psi_{2-}] + \frac{1}{4} \int dz e^{i2\Delta\theta} \Psi_{2-}^\dagger [a_{2-}, a_{2-}^\dagger] \Psi_{2-} , \\ [\downarrow | a_\downarrow^\dagger e^{-i2\Delta\theta} a_\uparrow | \uparrow] &= \frac{1}{4}[\Psi_{1+}^\dagger e^{-i2\Delta\theta} z \Psi_{1+}] + \frac{1}{4} \int dz e^{-i2\Delta\theta} \Psi_{1+}^\dagger [a_{1+}, a_{1+}^\dagger] \Psi_{1+} \\ &+ \frac{1}{4}[\Psi_{1-}^\dagger e^{-i2\Delta\theta} z \Psi_{1-}] + \frac{1}{4} \int dz e^{-i2\Delta\theta} \Psi_{1-}^\dagger [a_{1-}, a_{1-}^\dagger] \Psi_{1-} \\ &+ \frac{1}{4}[\Psi_{2+}^\dagger e^{-i2\Delta\theta} z \Psi_{2+}] + \frac{1}{4} \int dz e^{-i2\Delta\theta} \Psi_{2+}^\dagger [a_{2+}, a_{2+}^\dagger] \Psi_{2+} \\ &+ \frac{1}{4}[\Psi_{2-}^\dagger e^{-i2\Delta\theta} z \Psi_{2-}] + \frac{1}{4} \int dz e^{-i2\Delta\theta} \Psi_{2-}^\dagger [a_{2-}, a_{2-}^\dagger] \Psi_{2-} . \quad (47) \end{aligned}$$

Alternative Analysis of the Probability Amplitudes

In alternative to the above method, the probability amplitudes of the joint observation of the two photons, it could be convenient to adopt the “qbit” operators

$$\begin{aligned} [\uparrow | a_\uparrow^\dagger a_\uparrow | \uparrow] &= [\Psi_\uparrow^\dagger z \Psi_\uparrow] + [\uparrow | [\hat{a}_\uparrow, \hat{a}_\uparrow^\dagger] | \uparrow] , \\ [\downarrow | a_\downarrow^\dagger a_\downarrow | \downarrow] &= [\Psi_\downarrow^\dagger z \Psi_\downarrow] + [\downarrow | [\hat{a}_\downarrow, \hat{a}_\downarrow^\dagger] | \downarrow] . \end{aligned} \quad (48)$$

and

$$\begin{aligned} [\uparrow | a_\uparrow^\dagger e^{i2\Delta\theta} a_\downarrow | \downarrow] &= [\Psi_\uparrow^\dagger e^{i2\Delta\theta} z \Psi_\downarrow] \\ &+ \frac{1}{2} [0 | e^{i2\Delta\theta} ([\hat{a}_\uparrow, \hat{a}_\uparrow^\dagger] + [\hat{a}_\downarrow, \hat{a}_\downarrow^\dagger]) | 0] \\ [\downarrow | a_\downarrow^\dagger e^{-i2\Delta\theta} a_\uparrow | \uparrow] &= [\Psi_\downarrow^\dagger e^{-i2\Delta\theta} z \Psi_\uparrow] \\ &+ \frac{1}{2} [0 | e^{-i2\Delta\theta} ([\hat{a}_\downarrow, \hat{a}_\downarrow^\dagger] + [\hat{a}_\uparrow, \hat{a}_\uparrow^\dagger]) | 0] \end{aligned} \quad (49)$$

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